

My Design: Adaptive Technology
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Element K: Reflection on Design Project

Summary of project:

We started this project with hopes of dramatically improving the organization of the sensory room and even adding our own sensory product. We wanted to construct doors for the middle island first and then moving to doors for the cubbies so that files on the shelves around the room could be reorganized into the unused spaces where we wanted to add doors. We also planned on creating interchangeable tactile panels that could be added to the doors.

Completing these three mini projects proved to be more challenging and time-consuming than we thought, leading us to make the decision to focus on just the doors for the middle island. These doors would be complicated though, having the feature of being stored inside the middle cabinet as a pocket door when opened. Because the doors were large relative to the island they were covering, we also had to make these doors bi-fold, meaning they could fold onto itself, allowing it to be stored properly and removing any hazards that an open door close to the ground would bring.

Even when focusing on just the middle island doors, we found the construction process much more time consuming than we planned for it. In 'Element D', we planned on having our final doors ready to be installed March 12th, 2024. In reality, we weren't ready to install until about a month later at the end of April.

We were able to begin installation on May 15th, where we installed one of the doors. We came back May 29th to finish up, allowing us to install the second door, install 8 shelf supports, 4 shelves, and the 2 stationary panels which required an installation of 8 'L-shaped' brackets total.





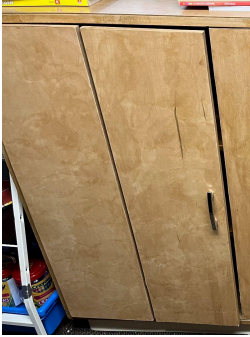
Reflection:

Overall, time management seemed to be the biggest issue, as well as making sure we had everything we needed. We seemed to face a lot of delays during construction, ranging from screw sizes to having to fix our broken rail that was essential to our pocket-fold system. We did a relatively good job in our research, but may have overlooked a lot of steps during construction. We definitely could have benefited from listing the specifics of construction so we could have a more accurate timeline, and communicate our plans better with our stakeholders. When we were ready to install our door, we faced some schedule conflicts with our stakeholder teacher from the elementary school with the sensory room and had to reschedule multiple times.

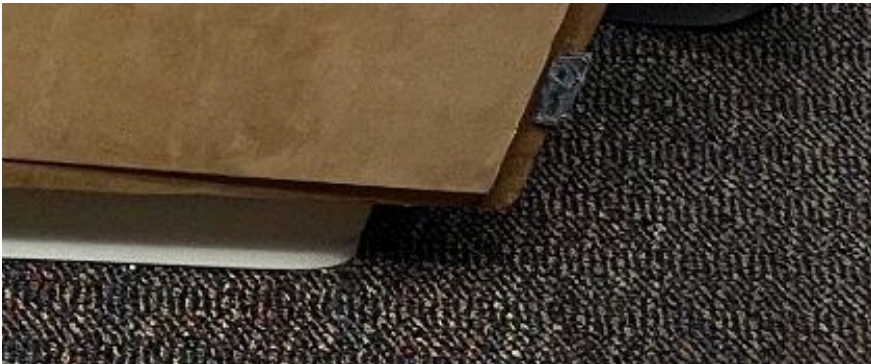
Overall, installation took a lot more time than we thought it would. The first time we went to install we could have been more prepared as there were some things that if we thought through we could have done on our own time, like pre-drilling holes and connecting the pocket rail to the vertical wood pieces. The second time we were much more prepared since we had a better idea of the installation process, but it still took us a long time. There wasn't much that could have been done besides having more prior experience since we were naturally slow. We were able to finish installing the project, with a key theme of underestimating the amount of time things take. Throughout the whole project, this has been a problem which could have been minimized with more attention to organization. If we did something like a run-through of installation before going, we would have had a better idea of the steps. I think we could have benefited from drawing these steps out ahead of time and assigning tasks so that when we got there we could stick almost to a schedule. This would give us and our stakeholders a better understanding of timing which was important since the classroom is used everyday.

All in all, we were able to complete our goal of improving the organization and utilization of space of the sensory room by installing doors to a largely unused middle island, and installing shelves so that things can be stored. There is now more space for sensory items to be added in spaces currently filled with files later down the road through other students' projects, and sensory items in general. Files will now be hidden by doors that complement the color of the rest of the sensory area.

One issue we noticed after installing was that the left door wouldn't close properly and it needed some additional support to keep it from bending outwards due to its bi-fold nature.



Additionally, we noticed that the right door was crooked in a way that makes the bottom of the door hit the bottom of the middle island, creating friction that halts the storing or opening of the door.



We also didn't plan on the doors sticking out as far as they did when being pocketed, but the main goal of having the doors out of the way is accomplished.

